

Question	Answer	Marks								
7(a)	$^{12}_7\text{N} \rightarrow ^{12}_6\text{C} + ^0_1\beta^+ + ^{(0)}_{(0)}\text{n}$	B1								
	beta-plus shown									
	neutrino shown	B1								
	symbols, nucleon numbers and proton numbers all correct	B1								
7(b)(i)	+1	B1								
7(b)(ii)	Lepton(s)	B1								
7(b)(iii)	<table><tr><td>flavour</td><td>charge / e</td></tr><tr><td>up / u</td><td>$-\frac{2}{3}$</td></tr><tr><td>up / u</td><td>$-\frac{2}{3}$</td></tr><tr><td>down / d</td><td>$(+)\frac{1}{3}$</td></tr></table>	flavour	charge / e	up / u	$-\frac{2}{3}$	up / u	$-\frac{2}{3}$	down / d	$(+)\frac{1}{3}$	
	flavour	charge / e								
	up / u	$-\frac{2}{3}$								
	up / u	$-\frac{2}{3}$								
	down / d	$(+)\frac{1}{3}$								
3 correct quark flavours	B1									
Charge on anti-up quark $-\frac{2}{3}(\text{e})$	B1									
Charge on anti-down quark $(+)\frac{1}{3}(\text{e})$	B1									